

The Hilbert Substrate Framework III: Bidirectional Links and Minimal Gauge Registers

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Abstract

This paper isolates an algebraic requirement for *bidirectional links* in the Hilbert Substrate Framework (HSF): a link register must decompose as independent representation spaces $V \otimes \bar{V}$ to host commuting left and right endpoint actions. We show that this implies a minimal link dimension threshold $d_B \geq N^2$ for $SU(N)$ endpoint structure, with the minimal case $d_B = N^2$ supporting a full matrix algebra M_{N^2} generated by the endpoints. We further demonstrate an explicit composite-link Hamiltonian that is gauge invariant and admits a nonempty Gauss-law singlet sector via the $3 \otimes \bar{3}$ vertex pairing. Finally, we operationalize “finite bandwidth” as a poke→endpoint influence map whose singular spectrum quantifies the number of strong, independent influence directions available through a link interface in one step.

1 Context and motivation

HSF proposes that particles and interactions are emergent collective structures of a constrained Hilbert substrate rather than ontological primitives. This third installment builds on Parts I–II [1, 2] by focusing on the algebraic structure of *links*: the interface degrees of freedom through which subsystems exchange influence. Our aim here is deliberately narrow: isolate the *minimal register size and factorization* required for a link to behave as a true bidirectional interface with independent endpoint representation spaces.

2 Bidirectional links as tensor-factor interfaces

Definition 1 (Bidirectional link register). *A link register is bidirectional for $SU(N)$ endpoints if its Hilbert space factors as*

$$\mathcal{H}_{\text{link}} \cong V \otimes \bar{V}, \tag{1}$$

where the left endpoint action is $\rho(g) \otimes I$ and the right endpoint action is $I \otimes \bar{\rho}(g)$, so that all endpoint generators commute: $[L^a, R^b] = 0$.

This definition expresses “endpoint ↔ endpoint” influence transmission as independent tensor factors, not merely two algebras embedded in a single small space.

Table 1: Constraint-driven factorization threshold experiment (script: `constraint_driven_factorization_v3.1.py`). For $SU(N)$, a uniform left embedding $L^a = T^a \otimes I_m$ exists only if $d_B = Nm$. The commutant is $I_N \otimes M_m$, which can host an independent $SU(N)$ right endpoint iff $m \geq N$, i.e. $d_B \geq N^2$. “NS+NF” counts trials that achieved commuting endpoints (NS proxy) and whose generated algebra filled M_{d_B} (NF proxy).

N	d_B	divisible	m	$\text{comm} \supset \text{su}(N)$	alg dim	NS+NF	GAUGE	
2	2	yes	1	no	0/4	0/6	0/6	
2	3	no	-	no	-	0/3	0/3	
2	4	yes	2	YES	16/16	6/6	6/6	MIN
2	5	no	-	no	-	0/3	0/3	
2	6	yes	3	YES	20/36	0/6	6/6	
3	3	yes	1	no	0/9	0/6	0/6	
3	6	yes	2	no	36/36	0/6	0/6	
3	7	no	-	no	-	0/3	0/3	
3	8	no	-	no	-	0/3	0/3	
3	9	yes	3	YES	81/81	6/6	6/6	MIN
3	12	yes	4	YES	90/144	0/6	6/6	

3 Minimal dimension threshold from commutants

Let d_B denote the link Hilbert dimension. Suppose a uniform $SU(N)$ left action is embedded as

$$L^a = T^a \otimes I_m \quad \text{on} \quad \mathcal{H}_{\text{link}} \cong \mathbb{C}^N \otimes \mathbb{C}^m, \quad (2)$$

so $d_B = Nm$. The commutant is $I_N \otimes M_m$, which can host an independent $SU(N)$ right action iff $m \geq N$, i.e. $d_B \geq N^2$. The minimal bidirectional register is therefore $d_B = N^2$ and corresponds to $\mathcal{H}_{\text{link}} \cong V \otimes \bar{V}$ with $\dim V = N$. We verify this threshold computationally for $SU(2)$ and $SU(3)$ using a corrected structure-constant projection test (Table 1).

4 Existence: composite-link gauge invariance and Gauss singlets

Beyond the threshold, we show an explicit composite-link construction for $SU(3)$ with $d_B = 9 = 3 \otimes 3$ and conjugate vertex pairing $3 \otimes \bar{3}$ at each endpoint. Using the script `gauge_test_focused.py`, we construct a two-leg coupling Hamiltonian and Gauss generators such that gauge invariance holds to numerical precision, and the Gauss subspace is nonempty (dimension ≈ 1). In the referenced run we obtain $\max_a \|[H, G^a]\| \approx 4.92e - 16$ and $\dim \ker(G^2) = 1$.¹

5 Finite bandwidth as an influence map

We avoid interpreting bandwidth as “bits per second.” Instead, we define a *poke*→*endpoint* influence map: apply a basis of small local perturbations to site A, evolve one step under the coupling, and measure induced changes in endpoint observables on the link. The singular values of this response matrix quantify how many independent influence directions transmit strongly through the interface in one step. Table 2 reports the top singular values for the composite-link conjugate model used above (script `bandwidth_profile_poke_map.py`).

¹See `repro/gauge_test_focused.20260222.113707.json` for the full record.

Table 2: Finite bandwidth operationalization as a poke→endpoint influence map. We apply a small unitary “poke” along each SU(3) generator on site A, evolve one step under the composite-link coupling, and record changes in expectations of left-endpoint observables. The top singular values of the resulting response matrix quantify how many independent influence directions pass strongly through the endpoint interface in one step.

Top singular values ($dt = 0.1$, $\eta = 0.1$):
0.00556717, 0.00402163, 0.00354176, 0.003359, 0.000163286, 7.55376e-05, 4.64137e-05, 1.20897e-06

6 Discussion

The results isolate a clean algebraic bottleneck: a bidirectional link register cannot be a single N -dimensional space if it must host independent left and right SU(N) endpoints. Instead, the minimal register is N^2 and naturally supports the $V \otimes \bar{V}$ factorization needed for commuting endpoints and conjugate pairing singlets. This provides a concrete target for implementing “finite bandwidth” and “no-refolding” as selection pressures that disfavor slack multiplicity ($m > N$) and encourage committed interfaces.

A Working Definitions of the Four Constraints

B HSF Constraints — Working Definitions (Project Info)

This page is a **living glossary** for the four core HSF constraints. Each definition has:

- **Core statement** (what it forbids)
 - **Operational test** (how we recognize it in sims)
 - **What it tends to select** (the pressure it applies)
 - **Failure modes** (how we can fool ourselves)
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B.1 1) No-signaling

B.1.1 Core statement

Influence cannot propagate everywhere at once. A change localized to one subsystem must affect other subsystems only through an allowed chain of interactions.

B.1.2 Operational test (in our current tooling)

- In an “endpoint/link” view: left and right endpoint actions on a link should be **independent interfaces**. A strong proxy is **commutativity** of endpoint actions on the link DOF:
- $\|[L^a, R^b]\|$ small for all a, b .
- In a lattice/graph view: couplings are restricted to an **interaction graph**; no all-to-all instantaneous updates.

B.1.3 What it tends to select

- **Interface structure**: information moves via boundary degrees of freedom.
- Often yields **commuting subalgebras** when the model has enough DOF.

B.1.4 Failure modes / caveats

- Commutativity alone is **too weak**: it allows reducible “hide-in-a-corner” solutions where most DOF are inert.
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B.2 2) No-forgetting

B.2.1 Core statement

The substrate cannot erase information. Evolution preserves distinguishability (reversibility), so histories remain recoverable in principle.

B.2.2 Operational test

- Dynamics is **unitary / reversible** (or an isometry on the relevant closed system).
- Practically: invariants (spectra, norms) are preserved under allowed evolution.

B.2.3 What it tends to select

- **Conserved structures** and persistent sectors.
- Promotes “memory” as **correlation structure**, not stored classical records.

B.2.4 Failure modes / caveats

- Open-system reductions (partial tracing) can look like forgetting unless we track the full Hilbert space.
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B.3 3) No-refolding

B.3.1 Core statement

Once structure (matter/excitations) exists, you cannot arbitrarily re-factor or rewire the substrate without paying a physical cost. Existing excitations constrain dynamically reachable configurations.

B.3.2 Operational test (candidate forms)

We track this in two nonexclusive ways (we’ll choose as the project stabilizes): 1. **Factorization stability**: once a link behaves like $V \otimes \bar{V}$, that tensor split is not freely mutable. 2. **Slack collapse**: large “do-nothing” commutants are disfavored because they represent refolding freedom.

B.3.3 What it tends to select

- ****Committed interfaces****: links become genuine bidirectional registers rather than arbitrary algebras inside a single space.
- Suppresses gauge-like “degeneracy solutions” that leave most DOF inert.

B.3.4 Failure modes / caveats

- If we only enforce no-signaling, we can accidentally allow massive refolding slack (large commutants) while still having commuting endpoints.
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B.4 4) Finite bandwidth

B.4.1 Core statement

A link cannot carry unlimited simultaneous influence. The interface has limited capacity to mirror/mediate changes between subsystems.

B.4.2 Operational test (HSF-native, no ‘bits/sec’ needed)

We treat bandwidth as a ****structural bottleneck**** on the ***operator channels*** that can pass through a link in one step.

Concrete proxy options (we can standardize one):

- ****Channel rank profile****: build an influence map $\Phi_A \rightarrow L$ from a basis of “pokes” on A to induced changes on the link endpoint, then inspect singular values.
- “Finite bandwidth” only a limited number of singular directions are significant.
- ****Active-operator budget****: only a limited number of coupling directions (in a fixed operator basis) may be large simultaneously.
- ****Spectral spread bound****: constrain $\|H_{\text{coupling}}\|$ / spectral width so influence can’t be arbitrarily strong across all channels at once.

B.4.3 What it tends to select

- ****Compression/hygiene****: the system learns to represent influence using a limited set of stable interface modes.
- Encourages genuine $V \otimes \bar{V}$ links when two independent endpoints must coexist under capacity limits.

B.4.4 Failure modes / caveats

- If bandwidth isn’t implemented explicitly, optimizers may find degenerate commuting solutions that underuse DOF (large commutants).
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C Cross-constraint notes

C.1 Bidirectional link definition (algebraic)

A “real gauge-like bidirectional link” means:

- $\mathcal{H}_{\text{link}} \cong V \otimes \bar{V}$ as independent tensor factors
- endpoint actions act as $\rho(g) \otimes I$ and $I \otimes \bar{\rho}(g)$ (automatically commuting)

A sharp witness for $d = N^2$:

- $\dim \text{Comm}(L) = N^2$
- $\dim \text{Comm}(R) = N^2$
- $\dim \text{Comm}(L, R) = 1$

C.2 Singlet-admitting vertex pairing

For $\text{SU}(3)$:

- $3 \otimes \bar{3}$ contains a singlet: $3 \otimes \bar{3} = 1 \oplus 8$
- $3 \otimes 3$ does not.

So Gauss-kernel nonemptiness naturally points to ****conjugate pairing**** at vertices.

D What we’ve learned so far (summary anchors)

- A single qutrit link ($d = 3$) can support clean $\text{su}(3)$ endpoint algebras but cannot host two commuting endpoints (the “48 wall”).
 - A composite link ($d = 9 = 3 \otimes 3$) can host a gauge-invariant Hamiltonian and, with conjugate pairing, a nonempty Gauss kernel.
 - No-signaling alone (implemented as “make endpoints commute”) does not force $V \otimes \bar{V}$ factorization or Gauss-kernel emergence; it admits reducible high-commutant solutions.
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E To decide next

We still need to standardize one operational definition of ****finite bandwidth**** in the codebase (rank-profile, active-operator budget, or spectral bound), because that’s the most likely missing pressure that collapses commutant slack and stabilizes true bidirectional links.

References

- [1] Ben Bray. The hilbert substrate framework. Manuscript (PDF), n.d. Project paper I.
- [2] Ben Bray. The hilbert substrate framework ii. Manuscript (PDF), n.d. Project paper II.